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Forecasting Tree Water Deficit in Switzerland

DAS Capstone

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Abstract

Tree water deficit (TWD) is an important indicator of tree water status and drought stress, derived from dendrometer measurements of stem variation. Despite the availability of large-scale datasets such as TreeNet, systematic approaches for forecasting TWD remain limited.

In this study, we formulate TWD prediction as a time series forecasting task using TreeNet data from 2020 to 2023. We compare autoregressive and non-autoregressive settings across multiple models, including ridge regression, long short-term memory (LSTM), and a Transformer encoder. We further investigate training strategies such as teacher forcing and rolling loss to improve multi-step forecasting.

Results show that autoregressive models leveraging past TWD values outperform non-autoregressive approaches, achieving high accuracy for next-day predictions. However, performance degrades with longer horizons due to error accumulation. Among all models, the Transformer with rolling loss provides the most robust performance for 7-day forecasting and shows improved generalization under distributional shifts, although all models struggle with out-of-distribution data.

This work highlights the effectiveness of machine learning for TWD forecasting and emphasizes the importance of model design and training strategies for robust multi-step prediction.

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Introduction

Real-time monitoring of drought stress in Europe’s temperate forests has become an important topic due to increasingly frequent droughts in Central Europe in recent years. Predictions of extreme climate drought events are necessary to support agricultural planning and biodiversity policy management [1].

At the individual tree scale, in-situ observations of stem expansion and contraction offer a direct and reliable proxy for assessing both growth dynamics and water status in many European temperate tree species. Tree water deficit (TWD) is one of the common indicators of stress due to water shortage for an individual tree. It leverages the biological facts that trees evaporate and lose water through leaves or needles and hence shrink their tree stems during the day, and re-expand their trunks at night to refill with water. In the absence of sufficient water, the stem shrinks. TWD is calculated as the difference between the current stem size and the historical maximum stem size. Sustained positive TWD values indicate insufficient water availability and can therefore serve as an indicator of drought stress and soil moisture conditions. TWD values can be measured systematically, near-real-time with a device called point dendrometer that is mounted on the stem wood at approximately 1.5m height [1, 2].

Switzerland’s TreeNet, established in 2010, is a nationwide forest monitoring and research network designed to collect and process high-resolution dendrometer measurements across broad temperature and precipitation gradients. The network comprises approximately 420 trees from 13 species distributed across 61 sites throughout Switzerland. TreeNet continuously records dynamic stem size variations with micrometer precision at 10-minute intervals. These measurements are automatically transmitted, processed, and stored on a central server, resulting in a dataset exceeding 100 million observations. This extensive dataset offers substantial potential for environmental applications, including the prediction of tree water deficit and the assessment of drought stress[2].

Despite the extensive temporal and spatial coverage of the TreeNet dataset, systematic efforts to forecast tree water deficit (TWD) remain limited. In this study, we leverage a subset of TreeNet data collected between 2020 and 2023 and formulate the problem as a time series forecasting task. Machine learning approaches are applied to predict TWD over a fixed forecast horizon.

Specifically, we investigate both autoregressive and non-autoregressive forecasting settings. In the autoregressive setting, predictions at each time step depend on previously generated outputs (i.e., past TWD values), whereas in the non-autoregressive setting, predictions are generated independently without relying on prior outputs (Figure 1.1)[3, 4].

To explore suitable model architectures for this task, we implemented and compared several machine learning approaches, including a simple linear model (ridge regression), a more complex neural network model based on long short-term memory (LSTM), and a Transformer model using its encoder architecture. Training strategies such as teacher-forcing and rolling-loss were applied to improve model performance.

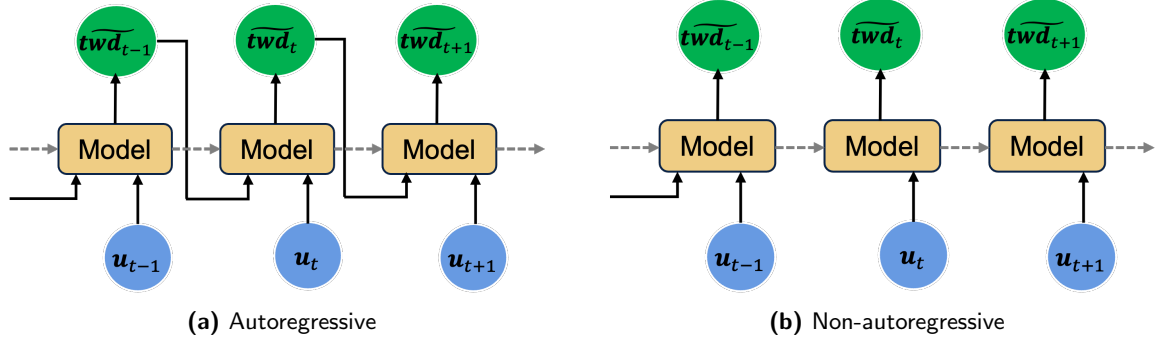


Figure 1.1: Schematics of forecasting tree water deficit values with autoregressive and non-autoregressive models. u indicates other temporal features being fed into the model at each time step, $\widehat{twd}$ indicates the tree water deficit value at each time step.

Methods

2.1 Data and Data processing

The tree water deficit dataset is a subset of the tstoy04 (TreeNet) spatio-temporal dataset and contains 42 tree series spanning from 2020-01-01 to 2023-12-31. Tree water deficit (TWD) values were recorded daily for 12 months and are accompanied with other exogenous temporal variable, primarily simulated by MeteoSchweiz. These include precipitation (kg m^{-2}), air temperature ($^{\circ}\text{C}$), wind speed (m s^{-1}), dew point ($^{\circ}\text{C}$), net shortwave radiation flux (W m^{-2}), and net longwave radiation flux (W m^{-2}). In addition, static variables e.g. tree species and their geographic information, were provided for each series. The latter comprises altitude, latitude, longitude, easting, and northing coordinates.

The Series from 2020-2022 was split into training, validation, and test set with a ratio of 3:1:1. The TWD series from 2023 was held out exclusively to evaluate model final performance.

We included the exogenous variables and static variables of the past 7 days as input features for the forecasting problem. We transformed the categorical attributes e.g. tree species and sites with one-hot encoders and standardized the numeric features e.g. the exogenous variables, to reduce the variance across features. $\text{Log}_2(1 + \text{TWD})$ transformation was applied to TWD values before modeling, a reverse transformation was then applied to model outputs, where negative values were clipped to 0 to ensure TWD values stay above zero.

In autoregressive setting, we included TWD values with exogenous variables for from past 7 days as feature inputs, whereas TWD values in the past days were left out in non-autoregressive setting. Static variables of tree species and tree sites were concatenated to the temporal features in both cases (Figure 2.1).

2.2 Models

To predict tree water deficit (TWD), three different model classes were evaluated: Ridge Regression, Long Short-Term Memory (LSTM) networks, and a Transformer encoder model.

Ridge Regression Ridge regression is a linear regression model with ℓ_2 regularization applied to the regression coefficients. The model estimates a linear relationship between the input features and the target variable while penalizing large coefficients to reduce overfitting. The optimization objective is

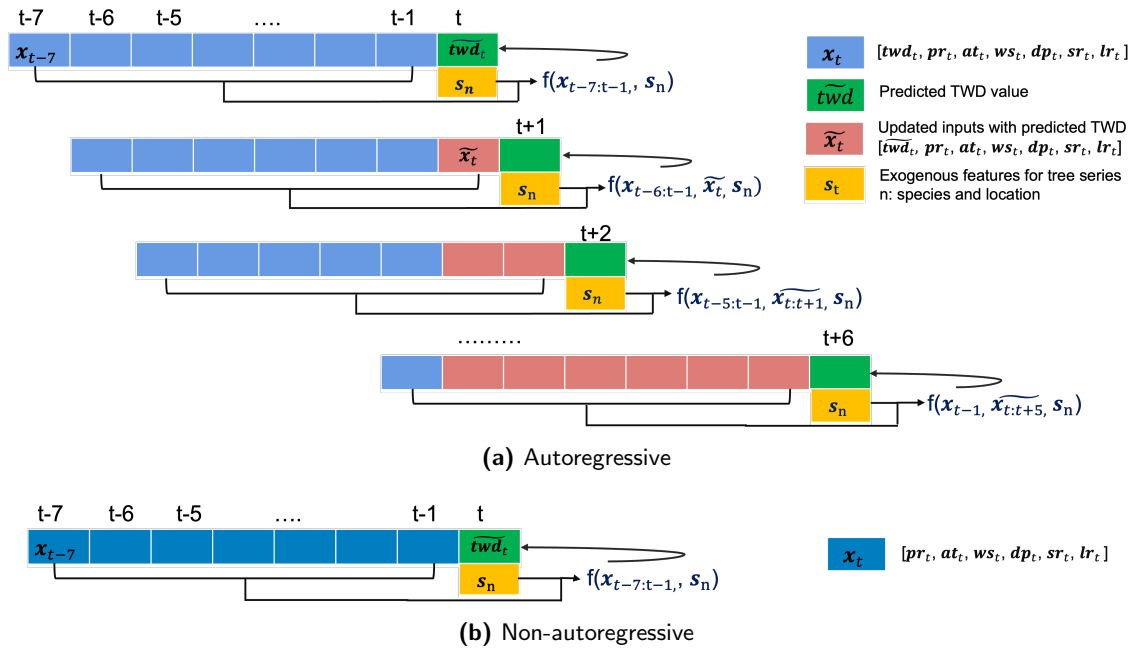


Figure 2.1: Feature inputs and outputs of models predicting 7-day-head TWD under autoregressive and non-autoregressive settings. $f(x)$: machine learning models. TWD: tree water deficit value; pr: precipitation (kg m^{-2}); at: air temperature ($^{\circ}\text{C}$); ws: wind speed (m s^{-1}); dp: dew point ($^{\circ}\text{C}$); sr: net shortwave radiation flux (W m^{-2}), lr: net longwave radiation flux (W m^{-2}). Suffix t indicates the time step, suffix n indicates tree series n . Note: LSTM and Transformer trained under the horizon set-up, which output the entire 7-day horizon in one forward pass, are not fully represented by the autoregressive schematic shown in (a).

$$\min_{\beta} \|y - X\beta\|^2 + \lambda \|\beta\|^2,$$

where λ controls the strength of the regularization[5]. Ridge regression serves as a strong baseline for time-series forecasting when lagged variables and exogenous features are included as predictors.

LSTM Long Short-Term Memory (LSTM) networks are a type of recurrent neural network (RNN) designed to capture temporal dependencies in sequential data. LSTMs use gated mechanisms (input, forget, and output gates) to control the flow of information through the network, allowing them to learn long-term dependencies while mitigating the vanishing gradient problem common in standard RNNs. In this study, the LSTM model processes sequences of past observations and exogenous variables to predict future TWD values[6].

Transformer Encoder The Transformer model is a sequence modeling architecture based on self-attention mechanisms rather than recurrence. In this work, only the encoder component of the Transformer architecture was used. The encoder consists of stacked layers of multi-head self-attention and feed-forward networks, allowing the model to capture dependencies between all time steps in the input sequence simultaneously. Self-attention enables the model to weigh the relative importance of different past observations when generating predictions[7]. In particular, the encoder consists of 3 attention layers with 4 heads in our setting.

2.3 Training Strategies and Loss Functions

Teacher Forcing Teacher forcing is a training strategy commonly used for sequence prediction models. During training, the model receives the ground truth value from the previous time step as input when predicting the next step. This stabilizes training and accelerates convergence because the model always conditions on correct historical information rather than its own predictions. However, during inference the true future values are not available, and the model must instead rely on its own previous predictions, which may lead to error accumulation[8].

Horizon Prediction In the horizon prediction strategy, the model directly outputs a multi-step forecast in a single forward pass. Instead of predicting one step at a time, the model outputs a vector of length seven representing the predicted tree water deficit values for the next seven days. This approach avoids recursive prediction and prevents error propagation across time steps, but requires the model to learn the relationships between multiple future time points simultaneously.

Rolling Loss The rolling loss strategy trains the model in an autoregressive manner while computing the loss across multiple forecast steps. After predicting the first step, the model's prediction is fed back as input to predict the next step, repeating this process for several future time steps. The training loss is then computed over the entire predicted sequence. This encourages the model to learn stable long-horizon predictions while accounting for the compounding errors that occur during recursive forecasting[8].

Teacher Forcing Loss For the teacher forcing training strategy, the model predicts the next time step while receiving the ground truth value from the previous time step as input. The training objective is computed using the Huber loss between the predicted and true values.

Given a prediction $\hat{y}_{t+1}$ and ground truth value y_{t+1} , the Huber loss is defined as

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & \text{if } |y - \hat{y}| \leq \delta \\ \delta (|y - \hat{y}| - \frac{1}{2}\delta), & \text{otherwise.} \end{cases}$$

The loss behaves like mean squared error for small errors and like mean absolute error for large errors, making it more robust to outliers[9]. The final training loss is computed as the mean Huber loss across batches.

Weighted Multi-step MAE Loss For the horizon and rolling-loss training strategies, a weighted multi-step mean absolute error (MAE) loss was used. Given a prediction horizon H , the model produces a sequence of predictions $\hat{y}_{t+1}, \dots, \hat{y}_{t+H}$, which are compared to the corresponding ground truth values $y_{t+1}, \dots, y_{t+H}$.

An exponentially decaying weight is applied to each prediction step so that earlier predictions receive higher importance. The weights are defined as

$$w_i = \frac{\exp(-\alpha i)}{\sum_{j=0}^{H-1} \exp(-\alpha j)}, \quad i = 0, \dots, H-1,$$

where α is the decay parameter controlling how quickly the weights decrease across the prediction horizon.

The loss for a single sample is computed as the weighted sum of the absolute prediction errors:

$$\mathcal{L} = \sum_{i=0}^{H-1} w_i |y_{t+i+1} - \hat{y}_{t+i+1}|.$$

During training, the final loss is obtained by averaging this quantity over all samples in the batch.

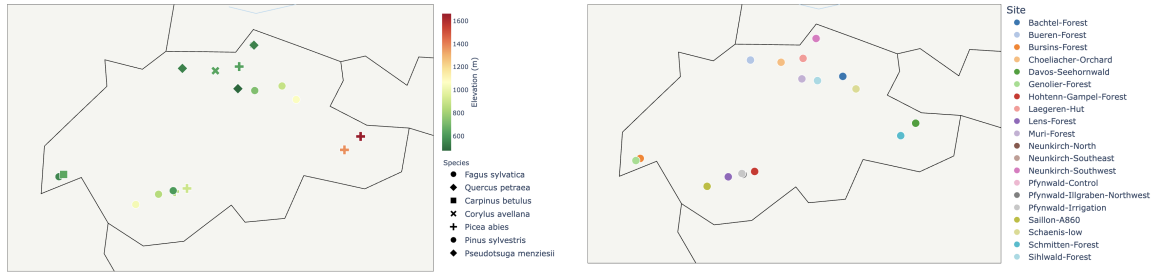
Model performance was evaluated using the root mean squared error (RMSE) and Pearson's coefficient (R^2). A brief model parameter selection was performed on training set with 4-fold cross-validation splits.

Results

3.1 Data Exploration

Effective modeling requires a thorough understanding of the underlying data. Therefore, we conducted exploratory data analysis using both visualization techniques and descriptive statistics to examine the 42 tree-level time series, assess data distributions, and identify temporal patterns.

As an initial step, we visualized the spatial distribution of tree locations and their corresponding species across Switzerland. The dataset comprises 42 time series from 7 tree species distributed across 20 forest sites, spanning a wide range of elevations—from 469 meters in Muri forest to over 1500 meters in Davos (Figure 3.1).



(a) Species map

(b) Site map

Figure 3.1: Species and sites of trees in 42 series included in our TreeNet subset.

Next, we explored the distribution of our targeted tree water deficit value in both training and final test set. Clearly, TWD values present a heavily zero-inflated distribution, slightly vary in training and final test set. In training set which spans from year 2020 to 2022, TWD values are mainly distributed from 0-4, with outliers spreading above 5, reaching 7. In the test set, outlier TWD values are reaching nearly 10, which might lead to a model performance gap between the training and test set (Figure 3.2). Subsequently, to identify informative predictors among the temporal covariates for TWD forecasting, we examined Pearson correlation coefficients between each covariate and TWD, as well as inter-correlations among covariates at the same time step (Figure 3.3). The results indicate

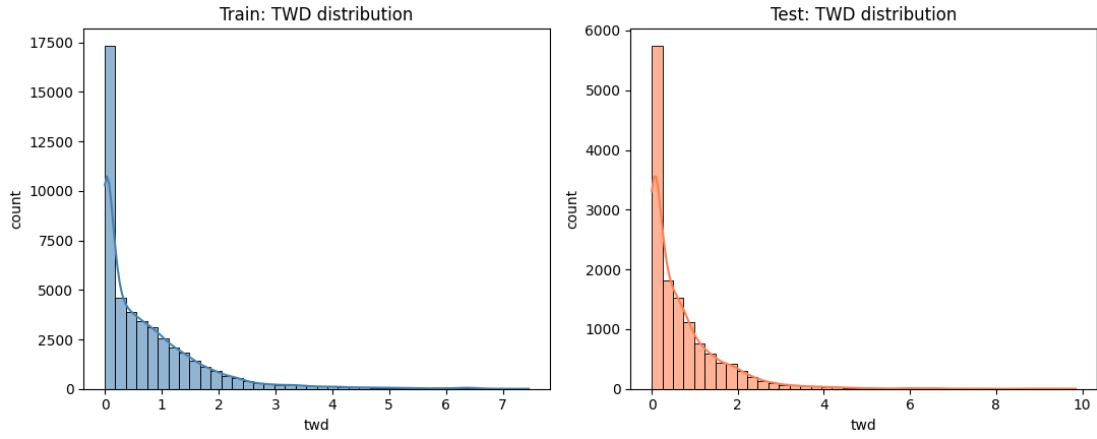


Figure 3.2: Distribution of tree water deficit values in training (2020-2022) and test set (2023).

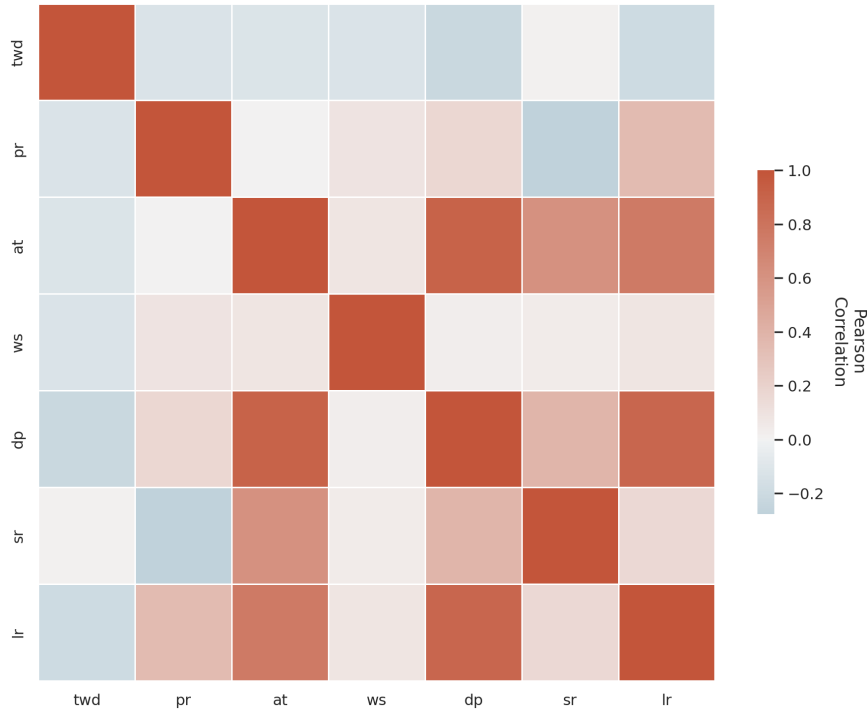


Figure 3.3: Correlation across TWDs and other temporal features. twd: tree water deficit value; pr: precipitation (kg m^{-2}); at: air temperature ($^{\circ}\text{C}$); ws: wind speed (m s^{-1}); dp: dew point ($^{\circ}\text{C}$); sr: net shortwave radiation flux (W m^{-2}), lr: net longwave radiation flux (W m^{-2}).

that none of the covariates exhibit strong correlations with TWD. Only weak negative correlations are observed between TWD and dew point, and between TWD and longwave radiation.

Notably, dew point, shortwave radiation, and longwave radiation show positive correlations with air temperature, reflecting their underlying thermodynamic relationships. In addition, we analyzed lagged correlations to assess whether covariates exhibit delayed effects on TWD. However, no strong lagged relationships were identified.

Lastly, we investigated the temporal dependence within the TWD series itself. As is standard in time series analysis, autocorrelation (ACF) and partial autocorrelation (PACF) were computed to examine the underlying temporal structure and potential periodicity.

Autocorrelation measures the Pearson correlation between the series and its lagged values, whereas partial autocorrelation quantifies the correlation at a given lag after removing the linear effects of intermediate lags[10].

Both ACF and PACF were evaluated for lags ranging from 0 to 50 days. The autocorrelation function exhibits a gradual decay from lag 0 to approximately 10 days, followed by a relatively stable region until around 15 days, after which the correlation approaches zero. This pattern suggests that the effective forecasting horizon for TWD lies within approximately 0 to 15 days, which aligns well with the common practice of short- to medium-term (1–2 weeks) forecasting.

In contrast, the partial autocorrelation function indicates a strong dependence at lag 1 and minimal contributions from subsequent lags, implying that the TWD value from the previous day is a highly informative predictor of the current value (Figure 3.4).

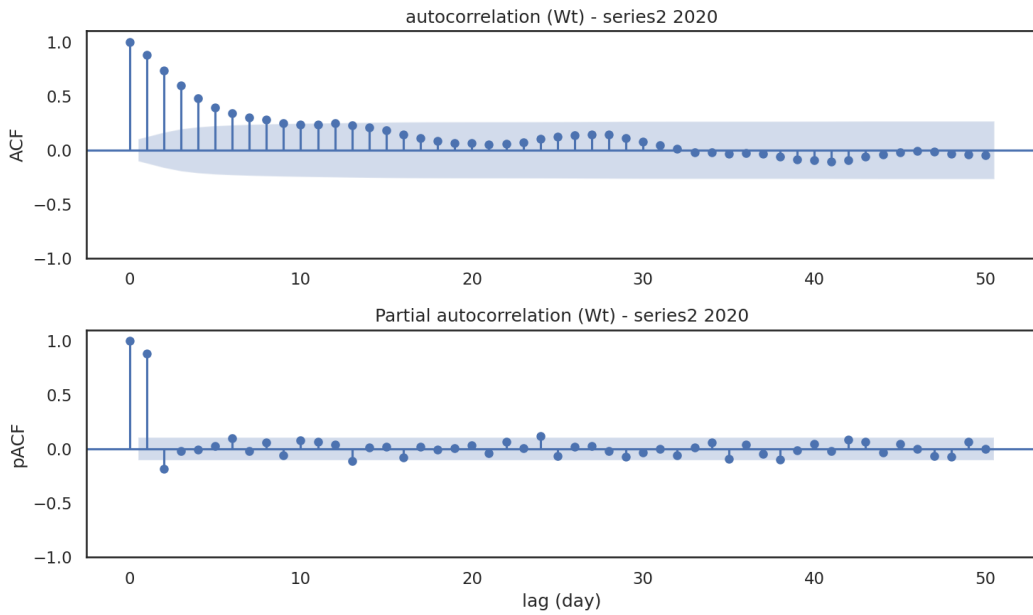


Figure 3.4: Autocorrelation and partial autocorrelation computed on series 2 of 2020 with lags from 0 to 50 days.

3.2 Forecasting with Autoregressive and Non-Autoregressive Models

After establishing a modeling foundation through the analysis of TWD distributions and autocorrelation patterns, we developed models with different methods and architectures to evaluate their forecasting performance.

As an initial step, we trained simple baseline models as a proof of concept, with the aim of understanding how model behavior differs under autoregressive and non-autoregressive settings. In the autoregressive setting, models were trained using a fixed input window of the past 7 days, including TWD values, to predict the TWD value at the 7th day ahead. During inference, the model generates the next-day prediction based on the input window, then recursively feeds its own predictions back as inputs to forecast subsequent time steps, until reaching the predefined horizon of 7 days (Figure 2.1).

In contrast, in the non-autoregressive setting, the model was trained to predict the next-day

TWD values using only exogenous features, excluding past TWD values. During inference, predictions are generated in a single forward pass without recursive updates (Figure 2.1).

We trained ridge regression models and evaluated their performance under both autoregressive and non-autoregressive settings.

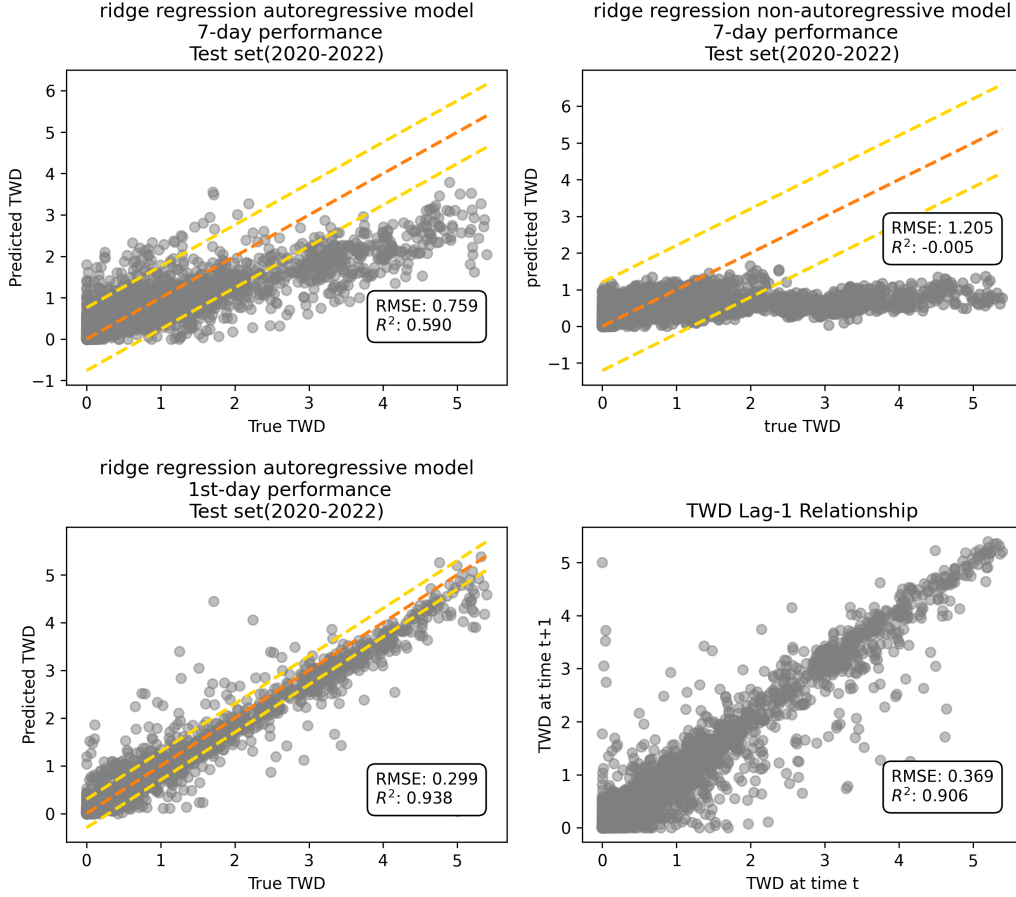


Figure 3.5: Ridge regression performance under autoregressive and non-autoregressive settings. Model performance was assessed on next-day and 7-day prediction with RMSE and R^2 score respectively.

As expected, the autoregressive model, which incorporates past TWD values, significantly outperforms the non-autoregressive counterpart. In particular, near-perfect performance is achieved for nex-day predictions in the autoregressive setting, with an R^2 score of 0.93, slightly surpassing the baseline of using the previous day’s TWD as the predictor. However, performance degrades rapidly as the forecast horizon increases, with a noticeable decline by the 7th day. This deterioration is likely due to error accumulation, as the model recursively conditions on its own predictions during rollout (Figure 3.5).

3.3 Comparing Performance of Autoregressive Models

We trained models with different architectures to predict TWD in an autoregressive setting, including ridge regression, long short-term memory (LSTM), and a Transformer-based model. Due to their distinct model structures, each approach is expected to capture different relationships between input features and the target TWD values.

Ridge regression, an ℓ_2 -regularized linear model, assumes a linear relationship between past features and future TWD [5]. In contrast, LSTM, a variant of recurrent neural networks, is specifically designed to model sequential data by propagating information through hidden states. In our setting, the LSTM processes the past 7 days of features as a sequence and leverages temporal dependencies to predict future TWD values.

Finally, we employ a Transformer model using only its encoder architecture. Originally developed for natural language processing, the Transformer utilizes attention mechanisms to model dependencies across input sequences. In this work, the attention mechanism is used to capture relationships across the past 7 days of TWD and exogenous features, enabling the model to learn complex temporal interactions for forecasting future TWD.

The models were trained with the same training set of tree series from year 2020-2022, and their performance was evaluated on the two test sets, respectively: 1. test set from the 2020-2022 series, split from the same set as the training set; 2. test set from the 2023 series, representing an out-of-distribution set in contrast to the training set.

In addition, several training strategies were explored to improve model performance under the autoregressive setting. In principle, models were trained to predict the next-day TWD value and then iteratively rolled out predictions over a fixed forecast horizon. Based on this next-step prediction framework, we implemented two training strategies: (1) teacher forcing and (2) rolling loss.

Teacher forcing is a widely used approach in which models are trained using ground truth values at each time step, thereby stabilizing learning during training. In contrast, the rolling loss strategy evaluates model performance over the entire forecast horizon by computing a weighted loss across the 7-day predictions, where each step is conditioned on the model’s own previous predictions. This approach better reflects inference-time behavior and has been shown to improve the robustness of sequence models, making it well-suited for autoregressive forecasting.

Furthermore, as an additional experiment, we developed horizon-based variants of the Transformer and LSTM models, which directly output a 7-day prediction vector in a single forward pass. This formulation simplifies the autoregressive forecasting process by reducing the need for recursive rollout.

Model	RMSE		R^2	
	1-day	7-day	1-day	7-day
Ridge regression (teacher forcing)	0.299	0.759	0.938	0.590
LSTM (teacher forcing)	0.284	0.702	0.944	0.650
LSTM-horizon	0.294	0.820	0.940	0.526
Transformer (teacher forcing)	0.340	0.714	0.920	0.638
Transformer-horizon	0.299	0.753	0.938	0.559
Transformer-rolling loss	0.338	0.686	0.921	0.665

Table 3.1: Comparison of model performance for 1-day and 7-day prediction horizons on the test set (2020-2022). Best results are highlighted in bold.

Model performance on test set 2020-2022 and 2023 is summarized in Tables 3.1 and 3.2 respectively. Root mean squared error (RMSE) and the coefficient of determination (R^2) were used to evaluate and compare model performance across two test sets. Overall, models achieved better performance on the 2020–2022 test set than on the 2023 test set, and

Model	RMSE		R^2	
	1-day	7-day	1-day	7-day
Ridge regression (teacher forcing)	0.869	0.734	0.250	0.466
LSTM (teacher forcing)	0.841	0.806	0.293	0.355
LSTM-horizon	0.813	0.797	0.342	0.369
Transformer (teacher forcing)	0.772	0.746	0.404	0.448
Transformer-horizon	0.753	0.739	0.435	0.457
Transformer-rolling loss	0.664	0.707	0.560	0.504

Table 3.2: Model performance comparison for 1-day and 7-day prediction horizons on the test set (2023). Best results are highlighted in bold.

consistently showed higher accuracy for next-day predictions compared to 7-day-ahead forecasts.

On the 2020–2022 test set, the LSTM model trained with teacher forcing slightly outperformed other models for next-day prediction, whereas the Transformer model trained with rolling loss achieved the best performance for the 7-day horizon. On the 2023 test set, the Transformer with rolling loss remained the most robust model, achieving an R^2 score of 0.504 for 7-day-ahead prediction, while other models exhibited substantially lower performance.

Notably, all models show degraded performance for next-day prediction on the 2023 test set compared to 2020–2022. This suggests that distributional shifts or out-of-distribution conditions in 2023 pose a challenge for model generalization at inference time.

In the sections below, model performance will be discussed in details.

3.3.1 Long Short-Term Memory

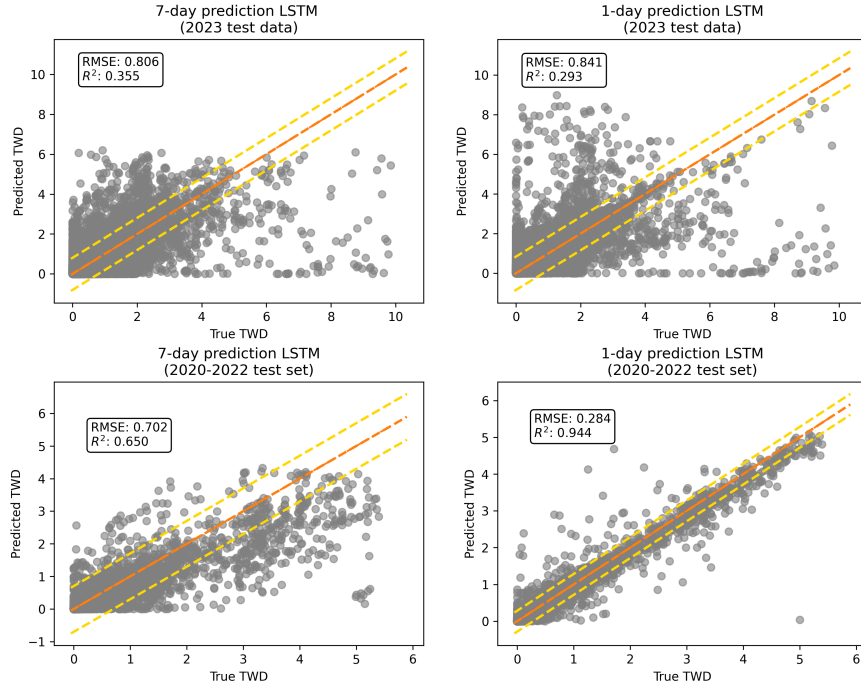
LSTM was first trained with a teacher-forcing strategy and subsequently adapted as an LSTM horizon model.

Both the teacher forcing and horizon-based models exhibit near-perfect performance for next-day TWD prediction on the 2020–2022 test set (Figure 3.6). Their performance declines for 7-day-ahead forecasts, with an RMSE of 0.702 and an R^2 score of 0.650, which remains acceptable. However, on the 2023 test set, model performance deteriorates substantially, with RMSE values increasing to approximately 0.8. In particular, the models fail to capture higher TWD values (e.g., above 6), indicating limited generalization under out-of-distribution conditions (see Discussion for further details).

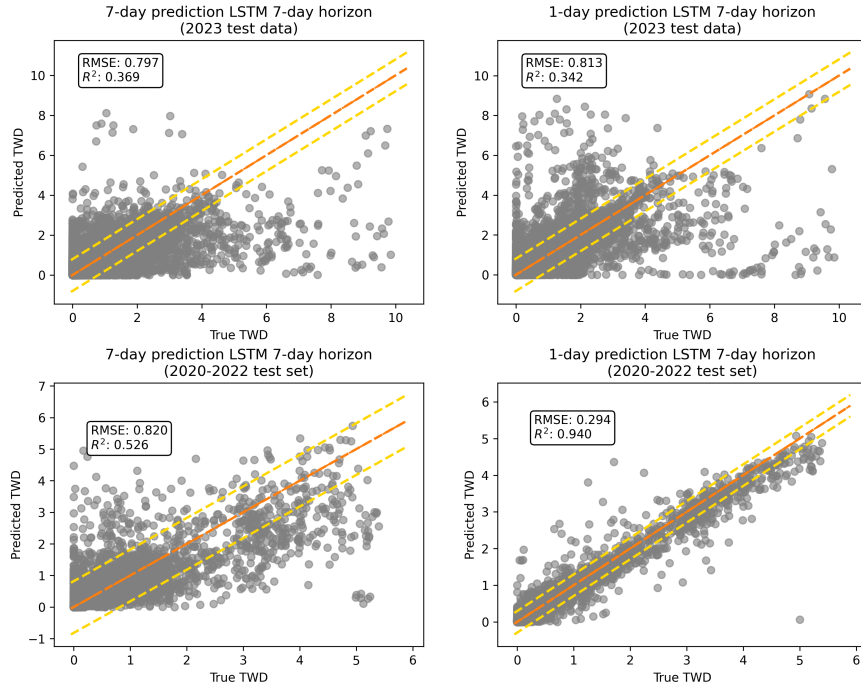
Notably, the teacher forcing model slightly outperforms the horizon-based model for 7-day-ahead prediction on the combined 2020–2023 test set. This suggests that the horizon-based formulation does not provide a clear advantage for the autoregressive forecasting task, especially given that both models perform poorly on the 2023 data.

Interestingly, on the 2023 test set, the RMSE for next-day predictions is higher than that for 7-day-ahead forecasts. An examination of the relationship between true and predicted TWD values indicates that model predictions tend to converge to a narrower range (approximately 0–5) during recursive rollout. As a result, long-horizon predictions exhibit smaller deviations from the mean range, leading to lower RMSE compared to next-day predictions.

3.3. Comparing Performance of Autoregressive Models



(a) LSTM (teacher forcing)



(b) LSTM-horizon

Figure 3.6: Comparison of LSTM models under teacher forcing and horizon settings. Performance of LSTM models predicting the 1st and 7th day TWD on the test set (2020-2022) and the test set (2023), respectively. Red dotted diagonal line shows $y = x$, whereas the yellow lines show $y = x \pm RMSE$.

3.3.2 Transformer

Similarly, we constructed a Transformer and a Transformer-horizon model trained with a teacher-forcing strategy. In addition, we trained a Transformer with a rolling-loss strategy.

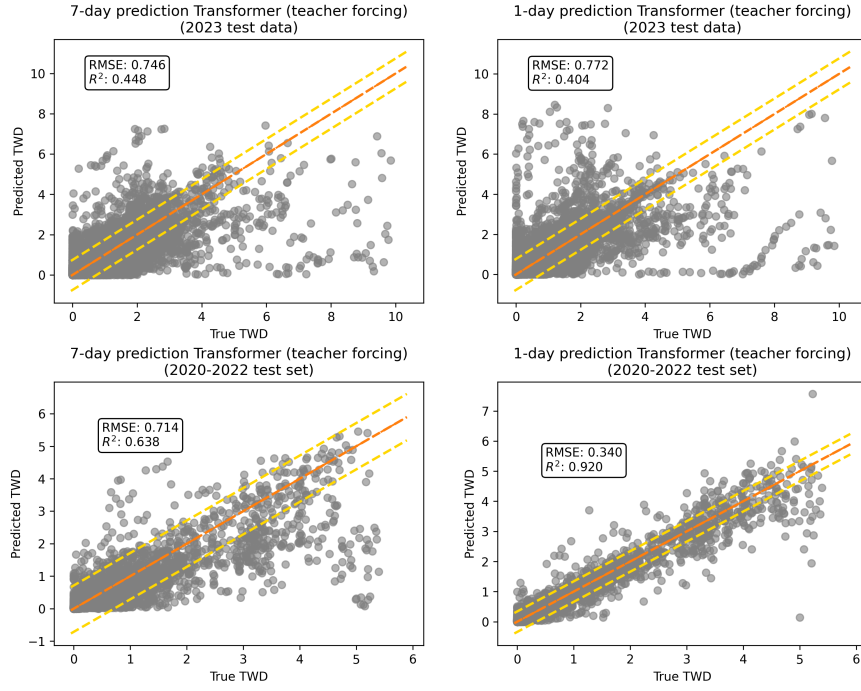
On the 2020–2022 test set, both the horizon-based and teacher-forcing Transformer models achieve strong performance for next-day predictions, while their accuracy decreases moderately for 7-day-ahead forecasts. However, a substantial decline in performance is observed on the 2023 test set, indicating a limited ability of the models to generalize to out-of-distribution TWD values (Figure 3.7).

To address the limitations of autoregressive training, we implemented a rolling loss strategy for the Transformer model, with the aim of improving performance under recursive forecasting.

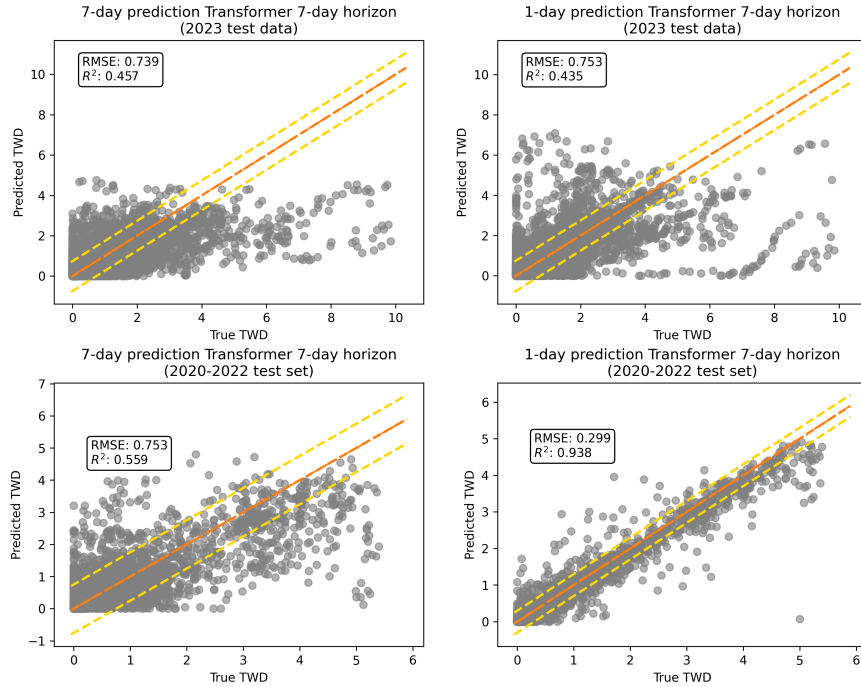
On the 2020–2022 test set, for the 7-day-ahead prediction task, the Transformer trained with teacher forcing exhibits a tendency to underpredict TWD values, as indicated by predictions lying below the identity line ($y = x$). In contrast, the Transformer trained with rolling loss slightly reduces this bias, producing predictions that are closer to the ground truth, although the overall patterns remain similar (Figure 3.8a).

Notably, compared to other models, the Transformer with rolling loss achieves improved performance on the 2023 test set, with R^2 scores of 0.560 and 0.504, and RMSE values of 0.664 and 0.707 for 1-day-ahead and 7-day-ahead predictions, respectively. While the model still struggles to accurately predict out-of-distribution TWD values, it demonstrates reliable performance within the range of 0 to 5 (Figure 3.8b).

3.3. Comparing Performance of Autoregressive Models



(a) Transformer (teacher forcing)



(b) Transformer-horizon

Figure 3.7: Comparison of Transformer models under teacher forcing and horizon settings. Performance of Transformer models predicting the 1st and 7th day TWD on the test set (2020-2022) and the test set (2023), respectively. Red dotted diagonal line shows $y = x$, whereas the yellow lines show $y = x \pm RMSE$.

3.3. Comparing Performance of Autoregressive Models

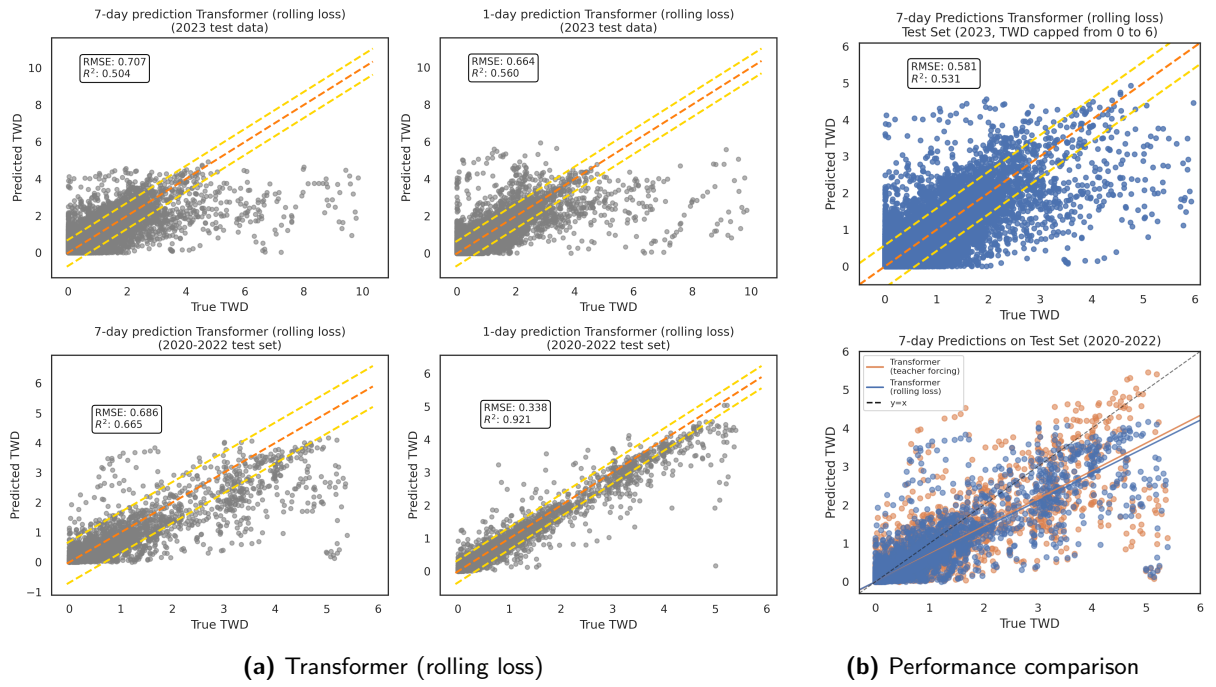


Figure 3.8: Performance of Transformer-rolling loss model. (a) Performance of Transformer-rolling loss model predicting the next-day and 7-day TWDs on test set (2020-2022) and test set (2023), respectively. Red dashed diagonal line shows $y = x$, whereas the yellow lines show $y = x \pm RMSE$. (b) Top: Transformer (rolling loss) performance on TWD ranges 0 to 6 on the test set (2023); Bottom: Transformer-rolling loss performance compared with Transformer (teacher forcing).

Conclusions and Discussion

We have explored forecasting the tree water deficit values with various machine learning models in autoregressive and non-autoregressive settings. Autoregressive models that included TWD values in the past as input features, unexpectedly outperform non-autoregressive model. Models showed better performance on predicting the next-day TWD values than on predicting the 7-day TWD values, and struggles to generalize to the out-of-distribution values. Among all autoregressive models we have constructed and compared the performance of, a Transformer trained with rolling loss strategy showed the best accuracy in forecasting TWD in seven days. However, as previously discussed, the best-performing model still failed to predict the out-of-distribution values in 2023 test set. Hence, it is of interest to further interrogate how the transformer model makes decisions in its attention head and what might differentiate the out-of-distribution values from the rest of the samples that might lead to the inaccurate predictions.

Firstly, we have extracted the attention weights across attention heads of the Transformer (rolling-loss) model (Figure 4.1) to reveal how correlated features of each day are with the features of other days, given attention weights are normalized dot products of queries and keys at different positions (days), indicating how similar the queries are to the keys, thus how big each key position contributes to final prediction of TWD, in other words, how much is the each query position attended to each key position. Passing a batch of 256 samples through the transformer models, we averaged the weights across the batch for each 3 multi-head attention layers of transformer rolling-loss and teacher-forcing models, respectively. The averaged attention weights of layer 3 in Transformer rolling-loss model, as the closest attention layer to the model output, showed that the queries are mostly attended to keys of the neighboring days, suggesting TWD and features from the last-day and next-day TWD are mostly correlated with the current-day TWD, consistent with our observations that the last-day TWD is a good predictor of next-day TWD. On the contrary, the Transformer teacher-forcing model showed on head 2,3 and 4 in layer 3, the queries, regardless of their positions, are attended almost undifferentiatedly to the past 4-day features before the forecast day, especially to the last-day feature, while head 3 in layer 3 shows attention of queries to all positions except for itself. The difference in attention weights between the two models confirms the effective training with rolling loss, suggesting that the model trained with rolling loss might have learned the neighboring dependence of TWD during training when the loss function is updated with the model's own predictions, whereas the teacher-forcing model might have only learned a mathematical function to minimize the next-day prediction error indicated by the high attention weight on the last day.

Next, we visualized locations and species of the out-of-distribution samples in test set

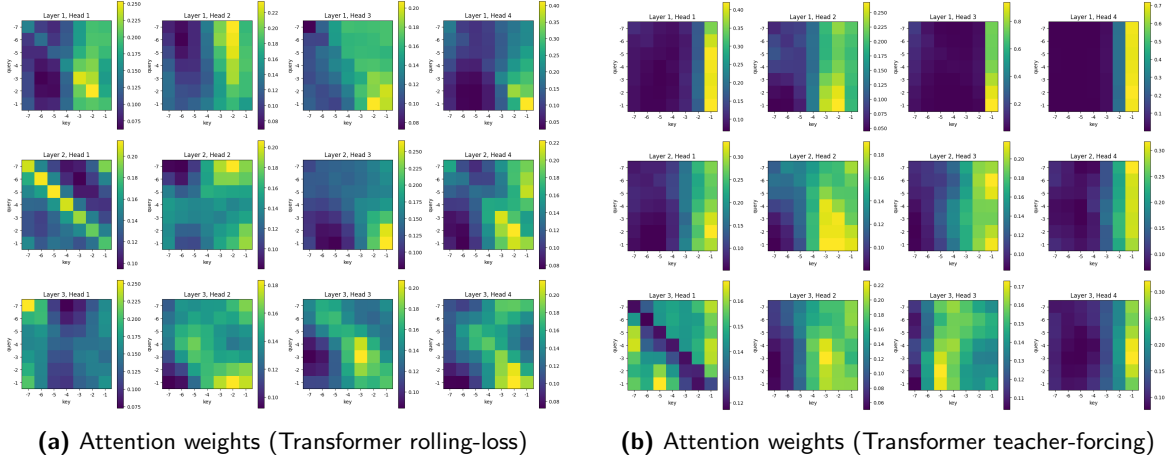


Figure 4.1: Transformer attention weights. Both Transformer models are composed of 3 multi-head attention layers and each layer contains 4 heads. layer 3 is the closest layer to the model output. Query and key positions indicate days of input features before the forecasted day.

2023, and attempted to explore the commonalities of TWD records where the Transformer rolling-loss model failed to predict accurately (Figure 4.2). These TWD values were collected from *Picea abies* from Davos-Seehornwald, Honteen-Gampel-Forest and Schmittens-Forest, *Psudotsuga menziesli* in Muri-Forest, which are also represented in the records with TWD less than 5. However, if we highlight the TWD outliers on the series together with predicted values, we observed the sudden surge of TWD values above 5 from relatively lower past TWD values, showing the gap between the past-day TWD and current-day TWD, which might be challenging for the model to extrapolate. Particularly, the series *Psudotsuga menziesli* in Muri-Forest showed the true TWD values stayed constantly high from January 2023 to May 2023, which might suggest abnormalities of tree or climate conditions during the period, necessitating further exploration (Figure 4.3).

Finally, several promising directions for future work can be considered. First, models could be trained on the 2023 data, which contains more extreme and outlier-like values, and subsequently evaluated on data from earlier years to further assess their generalization capabilities. Second, systematic hyperparameter optimization (e.g., grid search) could be employed to refine model configurations and potentially improve performance.

Lastly, expanding the dataset—either by incorporating additional years or increasing the number of monitored trees—could enhance model robustness and support more reliable forecasting performance.

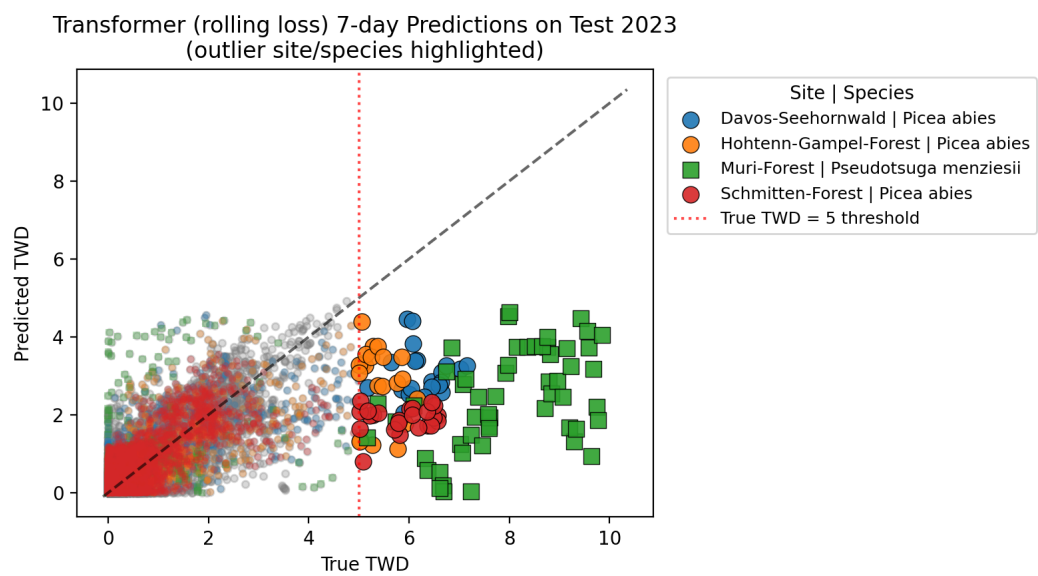


Figure 4.2: Species and locations of test set 2023 under predictions of the Transformer rolling-loss model, highlighting the records with true TWD greater than 5.

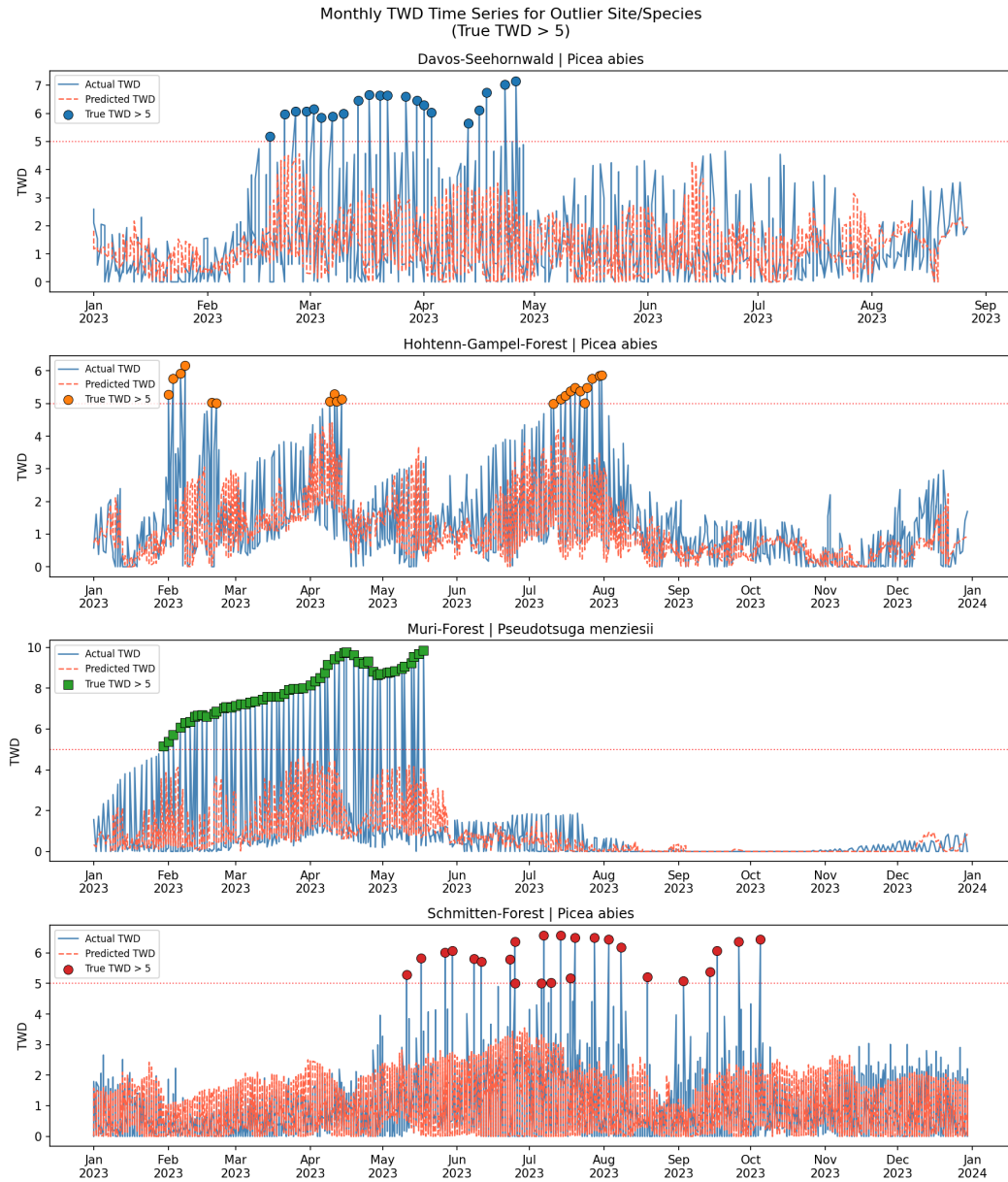


Figure 4.3: Species and locations of test set 2023 under predictions of the Transformer rolling-loss model, highlighting the records with true TWD greater than 5.

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